- Vendor Initial Reviews
    - Semantic Web PoolParty
        - Met: 3/21
        - Pros: Ability to share the results in a practical way externally, great for content applications. Tool & professional services are focused more on building applications using a semantic layer curated from the RDB world.
        - Cons: Bottom-up structure, firmly rooted in the RDB world of building apps on top of structures & tools,
        - Bottom Line: Not for us, feels like Semaphore redux
    - TopQuadrant TopBraidEDG
        - Met: 3/14 & 3/15
        - Pros: Governance built into the structure, Change history available, IRIs for instances: global settings available in EDG. Highly flexible based on how you want to reference different data, etc.
        - Cons: Implementation felt simplistic, TopBraid does not work with the instances – it's just the semantic models, Can't model with visuals – underlying code has to be manipulated, Graph Explorer gives simple, customized visualizations, similar to Protege. No filtering is available though.
        - Bottom Line: It could work but will require additional build-out on our part (no accelerators) and their professional services lack the experience with the scale and complexity we need.
    - Cambridge Semantics Anzo
        - Met: 3/20 & 3/21
        - Pros: Automated mapping, honest about capabilities & knowledgeable implementation staff, top-down & bottom-up modeling, Lots of internal functionality – dashboards, create APIs within the interface, view data/tables, modeling area. Geographical visualizations, Catalog, and models are auto-generated at data import. Models are based on metadata observed during import (schema etc) and can ingest external ontologies INCLUDING UML in XML format, but they need to be mapped to models generated automatically by Anzo, Data is not stored within Anzo, it is only brought into memory when a graph mart is created/actioned. Graph mart is the instructions used to build a knowledge graph, Validation and rules – need to be done judiciously, Metadata can be read from the data catalog. Their professional services (top guns) have delivered large robust semantic layers for large & complex companies. They have the chops.
        - Cons: Unsure how to integrate business users into the process,
        - Bottom Line: Top Contender, would like to move forward to the next steps.
    - StarDog
        - Met: 3/14, 3/18 & 3/20
        - Pros: Real application of knowledge graphs without needing heavy lift to transform, ability to utilize their RDF store should Marklogic be cut, implementation staff understand our difficulties, Catalog integrations

available (SD can be a catalog of catalogs), Inference computed at point of query, Query Profiler will automatically generate SQL–can help with tuning/optimization, Caching mechanism available for querying. Professional services are solid and they bring the experience we need for scale and complexity.
        - Cons: Cloud-only implementation, which may be a concern from a security and risk management perspective. Integration and performance may be a limiting factor as we build scale. Visualization is limited when managing taxonomies and working with hierarchies
        - Bottom Line: Top Contender; would like to move forward to next steps.
    - AbInitio
        - Reviewed by Sherri to be used in conjunction with Protege
        - Pros: It does look like we could manage glossaries & taxonomies in this tool relatively simply.
        - Cons: I do not have access to any documentation that shows me exactly what this would look like and creating anything in the POC requires someone else to approve it due to my permission settings. I don't like that all the created content is shared across the platform, so anyone with user abilities above mine could adjust or change any attributes or content I create, I'm not sure that their idea of an ontology is the same as mine and they aren't based on W3C standards. Governance may be a concern with this open semantic hub approach. This would require further discussion with the vendor. In addition, they do not provide any support for mapping to RDFs and other types of content other than the metrics & context gleaned from sources & structures used to deliver analytics. This strictly focuses on defining and maintaining hierarchies to support analytics and defining relationships via inference (Semantic Discovery) to terms culled from data structures. There are use cases for bi-directional integration with the SKL, but this would only support a subset of SKL. It can replace the Metrics Catalog and Metrics Engine. I do think Ab Initio can accelerate and scale our efforts to create semantically informed microservices, but the integration architecture is on hold for Corporate Data & Ecosystem.
        - Bottom Line: Could possibly work, but lots of work around required without a dedicated support team so launch would be difficult
    - Alation
        - Reviewed by Sherri to be used in conjunction with Protege
        - Pros:
        - Cons: DBT's semantic layer is primarily focused on mining context about the metrics lifecycle, improving reuse and search (finding the right metric). There are use cases for bi-directional integration with the SKL, but this would only support a subset of SKL. May be able to replace the Metrics Catalog.
        - Bottom Line:
- Next Steps for Short List

- Requirements List To Be Refined
  - Experience implementing formal knowledge engineering in companies
  - Examples of lifecycle covering taxonomies > ontologies > KGs > Context Mapping to Content
  - Experience mapping enterprise semantics to isolated / islands of semantics across data repositories, processes, and systems.
  - Guidance to ingest Metamodels and/or Catalogs to support reverse engineering and republishing from the enterprise semantic layer
  - Minimum Requirements
    - Compliance with ISO/NISO
    - Ability to model W3C ontologies (OWL)
    - Ability to model W3C vocabularies (SKOS)
    - Triples can be exported and used by other tools without large data manipulation effort
    - Simple integrations to import content from other resources
    - Simple integrations to export content to other resources like third-party RDF stores
    - Ability to create and manage concept-based schemes
    - Ability to create and manage terminology-based vocabularies such as thesauri and taxonomies
    - Ability to create and manage inference-bearing class-subclass hierarchies
    - Ability to allow different user communities to curate alternative sets of preferred terminology for a common conceptual base
    - Unique Identifiers (GUIDs and URIs/IRIs) for all concepts
    - Ability to view a concept's relationships as a flat list, a hierarchical list, or a graph visualization
    - Simple and advanced search modes
    - Ability to manage multiple project teams and assign users and taxonomies
    - Ability to support granular control of permissions for users on a per-project basis
    - Ability to share selected taxonomies with non-editorial stakeholders
    - Batch editing, archiving, and versioning tools
    - Multiple data formats for viewing, importing and exporting vocabularies
    - Import and export in multiple RDF serializations
    - import and export in CSV, TAB, and Excel file formats
    - User account management and role-based permissions
  - Would Like to Have
    - Tool agnostic URI/IRI generator
    - Ability to allow different user communities to curate alternative sets of preferred terminology for a common conceptual base

- Automatic prevention of term duplicates (disambiguation)
- Automatic inverse relationships
- Automatic prevention of circular references
- Support for poly-hierarchical structures
- Ability to import lists of candidate terms from spreadsheets and text files
- Ability to create new concepts without leaving the workflow of relationship editing
- Ability to browse alphabetically
- Ability to browse hierarchically and by visualization graphs
- Ability to create custom workflow and governance states to control the flow of work
- Ability to create management reports that can be shared and reused by other users
- Ability to track all editorial activities by activity type as well as date-time and UserID

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Note: if we are going to reference the list below, we may want to score the top 2 vendors

Capability	
FUNCTIONAL SECTION	
Vocabulary Management	Ability to perform text analysis and extract text
	Ability to support semantic associations
	Ability to assign attributes/properties to nodes and edges
	Ability to categorize edges
	Ability to mine new terms from imported RDFs, OWL structures
	Ability to assign weights to edges (esp. for rules)
	Ability to support Reasoning & Inference Rules (RIF) language (leveraging terms from SPIN or SHACL)
	Ability to instantiate math symbols (calcs, rules, etc.)
	Ability to construct consistent URIs and IRIs
	Ability to index with a universal identifier that we can rely upon for identifying all data (and things) separate from their labels & properties
Taxonomy Management	Ability to create SKOS standard vocabularies
	Ability to support auto tagging and semantic extensions
	Ability to organize entities into hierarchies
	Ability to build associated taxonomies
	Ability to manage taxonomies
	Ability to support direct and indirect inheritance
	Ability to support change management
	Ability to support version management

	Ability to manage workflow and control
	Ability to perform impact analysis
	Ability to publish or promote taxonomies
	Ability to push warning messages on moving or pruning nodes
Ontology Management	Ability to define patterns in ontologies
	Ability to evolve taxonomies into a KG
	Ability to conduct NLP supported search
	Ability to deprecate ontology without deletion
	Ability to use Declarative Query Language e.g. SPARQL, GraphQL, Cypher etc
	Ability to support ACID transactions
	Ability to perform augmented search
	Ability to navigate structured and unstructured data
	Ability to harmonize & query content from multiple sources
	Ability to support evolutionary development & engineering
	Ability to match/align ontologies
	Ability to match/align schemas
	Ability to match/align semantics
Relationship Management	Ability to support meta property (attributes, properties for relationships)
	Ability to represent event relations (n-ary support)
	Ability to extract information from entities and relations using inferred meanings
	Ability to extract information from entities and relations using disambiguation
Data Lifecycle Management	Ability to manage NULLs and duplicates in data
	Ability to transform data by using techniques like matching, fusing, merging etc
	Ability to support and enforce adherence to knowledge capture methodology (discovery – vocabulary – taxonomy - ontology)
	Ability to support and enforce adherence to implementation methodology (validation – mapping – query – measurement)
	Ability to capture inputs, processing, outputs across lifecycle
	Ability to support for impact analysis and simulation
	Ability to manage incremental and iterative development
Visualization Management	Ability to visualize data in tree form
	Ability to optimize number of nodes
	Ability to optimize weights on nodes

	Ability to provide for filtering
	Ability to manage and run NLP Query
	Ability to support for optimization / pathing
	Ability to provide for exploration
	Ability to support a multimodal graph db (RDF + LPG)
Model Capabilities	Ability to support transaction model (ACID)
Digital Twins	Ability to support Digital Twin-specific use cases
NON-FUNCTIONAL SECTION	
	Ability to scale performance linearly and horizontally
	Ability to support centralized and distributed graph DBs
	Ability to support SHACL constraints & rules construction
	Ability to support location-based intelligence
	Ability to support on-prem & off-prem (multiple clouds)
	Ability to include or accommodate AI extension
Performance	Ability to include or accommodate extension for data integration
	Ability to include or accommodate analytics and aggregation extension
	Ability to include or accomodate modelling and analysis extension
	Ability to include or accomodate extension for recommendation engine
	Ability to include or accomodate extension for sematic data integration
	Ability to include or accomodate extension for social network analysis
	Ability to support business continuity using backup/restore process
	Ability to support business continuity by transactions recovery process
	Ability to scale unlimited to store and manage large volume of data
	Ability to scale horizontally and veritically
Scalability	Ability to deploy with auto scaling in AWS/GCP
	Ability to support Kubernetes in AWS/GCP
	Ability to handle concurrent users(~50 users) searches
	Ability to provide API support for bi-directional formats like RDF, RDFS, OWL, XSD, SKOS, JSON
	Ability to provide security support using LDAP, RBAC
Integration (import/export)	Ability to support integration (bi-directional) between RDB & RDF using RDF2RDB Mapping
	Ability to support integration (bi-directional) between RDB & RDF using R2RML for RDFs

	Ability to support integration (bi-directional) between RDB & RDF using SQL Queries
	Ability to support integration (bi-directional) between RDB & RDF using SPARQL Queries
	Ability to support integration (bi-directional) between RDB & RDF using Virtual mapping (SQL2RDF)
	Ability to support integration (bi-directional) between RDB & RDF using D2R Optimization (for querying RDB)
Security	Ability to integrate with IdP providers to achieve enterprise SSO and authentication services.
	Ability to provide role-based access control.
	Ability to monitor, track, audit access control of the system.
	Ability to add additional owners and/or editing rights to assets.
	Ability to grant different levels of access to assets to control granular user access permissions.
Compliance	Ability to support GDPR, CCPA compliance requirements around data privacy such as accoutability, data minimization, lawful processing.
	Ability to comply with standards such as ISO/IEC-27001/27002, GSAM and FISMA requirements.
	Ability to support an audit trail of data access/modifications by User ID and by date/time
Availability	Ability to support active-active server configurations during cluster failover and disaster recovery.
	Ability to support 99.99% uptime SLA for operational support.
	Ability to provide fail-safe measure if db host fails
	Ability to store and backup assets in case of DR or Cluster Failover
Backup & Recovery	Ability to use centralized backup/recovery catalog across multiple databases
	Ability to support failover from Primary to Standby on unplanned primary DB failure
	Ability to support data replication
	Ability to support for add/remove of Standby/DR database without impacting Primary
Support	Ability to provide support 24/7

- External Review
  - https://www.stardog.com/news/awards/
  - https://cambridgesemantics.com/category/news/
  - https://www.gartner.com/reviews/market/active-metadata-management/ve

ndor/cambridge-semantics/product/anzo
    - https://www.gartner.com/reviews/market/cloud-database-management-systems/vendor/stardog/product/stardog
    - https://www.g2.com/products/stardog/reviews
    - https://www.g2.com/products/anzo/reviews#survey-response-7314061

Suggestion:

- Top-Down: Governance Team recommends the Semantic Modeling Tool with partnership from Business, not Technical folks. KE team is driving & facilitating the SKL
- Bottom Up: Governance Team recommends open standards for RDF stores PLUS IRI Generation & Preservation so we can leverage one or more RDFs (perhaps starting with initial footprint from Top-Down vendor)

Since we are proposing the 3-Tier Architecture for SKL:

- Semantic Knowledge Layer (SKL - Domain Fabric)
- Unified Data Model (RDF - Fabric)
- Event-Driven Architecture
    - Loading into RDFs
    - Orchestration
    - Discrete Services (Data, Analytics, AI, LLM)
- Learning
    - Recursive Training for Domain LLMs using Domain Ontologies
    - RAG Training for Properties & Relationships (varying levels of granularity within SKL)
    - Controlled Inferencing for exploration and impact (scenario extension - cached only)
- Loading
    - SOT (Metamodel, Catalog, Ab Initio, Collibra, Engram, etc.)
    - SORs (Data, Docs, Metrics, Analytics, AI, LLMs)